

M07 / STUDY COMPANION

M07 Quick Revision Book

Standalone Zero-to-Job-Ready Data Engineer 2026

Study, practise, check and explain.

A complete learning route

Start with the lesson video. Read the worked explanation beside it, pause to reproduce the example, then solve the supplied task. The books contain your study notes; keep your own work and results as evidence.

Use the protected answers after saving your first attempt. A changed retry shows what you can now do independently. There is no forced timer during learning.

One course. M00-M19. 264 lessons and 50 project phases.

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Tap a topic or use the PDF bookmarks. These numbers are the actual pages in this edition.

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Read this first

QUICK REVISION BOOK

Use this only after you have studied the learner book. Each lesson keeps the canonical language and core proof, plus one recall task. This is not a replacement for hands-on practice.

Coverage: 13 canonical lessons

Route: learner book -> practice -> revision -> closed-book test -> protected answers -> changed retry.

How to revise

Read the lesson card, close the page, explain it in your own words, then answer the recall task. If recall is weak, return to that lesson only - not the entire module.

DE07-01 | Verify PostgreSQL connection and database/schema context

60-second recall

What strong reasoning looks like A strong attempt proves context from the active PostgreSQL session, not from the pgAdmin browser tree or your memory. You should be able to distinguish server/cluster, database, schema and table. If `current_database()` is wrong, reconnect before DDL. If `to_regclass('table')` and `to_regclass('schema.table')` differ, explain that unqualified lookup is search-path dependent. Without opening the lesson, explain Verify PostgreSQL connection and database/schema context in 60-120 seconds. State the row/object grain, one realistic failure, one database-enforced rule (if applicable), and one indepe

[Refresh source: freeCodeCamp - Learn PostgreSQL Tutorial - Full Course for Beginners](#)

Recall task

Use raw/serving schemas with the same table name; predict three `search_path` resolutions before querying.

DE07-02 | Tables, data types and explicit DDL

60-second recall

What strong reasoning looks like Review your types by business meaning. Identifiers are not numbers merely because they contain digits. Exact currency-like values should not silently become approximate floats. A date is different from an event instant. Without opening the lesson, explain Tables, data types and explicit DDL in 60-120 seconds. State the row/object grain, one realistic failure, one database-enforced rule (if applicable), and one independent validation.

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Recall task

Design receipt header/line DDL with leading-zero business ID, exact money, date and one generated surrogate.

DE07-03 | Primary keys, foreign keys and constraints

60-second recall

What strong reasoning looks like Your constraint tests should include successful and rejected rows. Preserve at least one PK/UNIQUE, FK, CHECK and NOT NULL failure where relevant. If a CHECK alone was expected to reject NULL, revisit PostgreSQL CHECK semantics: an expression that becomes NULL is not false, so NOT NULL may also be required. Without opening the lesson, explain Primary keys, foreign keys and constraints in 60-120 seconds. State the row/object grain, one realistic failure, one database-enforced rule (if applicable), and one independent validation.

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Recall task

Receipt line composite key plus missing-parent/blank-text failure set; state what each rule still cannot prove.

DE07-04 | Transactions, DML and UPSERT

60-second recall

What strong reasoning looks like For transaction mastery, show state before the transaction, changed state inside it, and restored/committed state afterward. If an error caused "current transaction is aborted," the first error is the root cause; rollback or rollback to a savepoint restores usable transaction state. For UPSERT, name the conflict key and the row grain it represents. Without opening the lesson, explain Transactions, DML and UPSERT in 60-120 seconds. State the row/object grain, one realistic failure, one database-enforced rule (if applicable), and one independent validation.

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Recall task

Supplier current-state upsert + intentional two-write failure; explain why SCD2 cannot use overwrite semantics.

DE07-05 | Indexes and EXPLAIN evidence

60-second recall

What strong reasoning looks like A good index explanation begins with the query pattern and selectivity, not the index command. On tiny tables, Seq Scan can be the rational plan. EXPLAIN cost is planner cost, not milliseconds. EXPLAIN ANALYZE executes the statement. Compare the same business result before/after any index change. If you changed filters while comparing plans, your experiment is invalid. Without opening the lesson, explain Indexes and EXPLAIN evidence in 60-120 seconds. State the row/object grain, one realistic failure, one database-enforced rule (if applicable), and one independent validation.

[Refresh source: Postgres Conference - Query Optimizations and Where to Find Them: A Beginners Guide](#)

Recall task

Create dominant-vs-rare hub distribution and predict plan/index usefulness for each predicate.

DE07-06 | Operational vs analytical modeling and normalization

60-second recall

What strong reasoning looks like A strong model separates current operational facts from analytical reporting needs. In a flat order-line table, repeated customer attributes create update anomalies. 1NF/2NF/3NF language is useful only if you can connect it to practical dependencies. Normalizing customers/orders/order_items does not automatically preserve historical region. Without opening the lesson, explain Operational vs analytical modeling and normalization in 60-120 seconds. State the row/object grain, one realistic failure, one database-enforced rule (if applicable), and one independent

[Refresh source: Decomplexify - Learn Database Normalization \(1NF, 2NF, 3NF...\)](#)

Recall task

Supplier receipt flat export -> normalized operational model -> star; identify one update anomaly and one history question.

DE07-07 | Facts, dimensions and star-schema grain

60-second recall

What strong reasoning looks like Fact grain must be one unambiguous sentence. For the supplied model it is one completed order line. The core source controls are 8 completed lines, 5 completed orders and 1780.00. Repeating an order-level total/fee on each line creates fan-out even though each repeated value is individually correct. Region controls under the SCD2 history are East 660, South 600 and West 520. Without opening the lesson, explain Facts, dimensions and star-schema grain in 60-120 seconds. State the row/object grain, one realistic failure, one database-enforced rule (if applicable), and one independent validation.

[Refresh source: TechLake - DWH Tutorial 6: Star Schema in Dimensional Modeling](#)

Recall task

Receipt-line fact with one header freight amount; choose a safe modeling strategy and reconcile changed controls.

DE07-08 | Surrogate keys and referential integrity

60-second recall

What strong reasoning looks like The natural/business key answers which entity; the surrogate key answers which warehouse row/version. A surrogate primary key does not prevent two rows representing the same business entity unless a separate business/history rule does. In SCD2, multiple surrogate rows for one customer are intentional, so `UNIQUE(customer_id)` across the whole history is wrong. Without opening the lesson, explain Surrogate keys and referential integrity in 60-120 seconds. State the row/object grain, one realistic failure, one database-enforced rule (if applicable), and one independent validation

[Refresh source: freeCodeCamp - Learn PostgreSQL Tutorial - Full Course for Beginners](#)

Recall task

Product SCD2 dimension where one business product legitimately has several surrogate versions.

DE07-09 | Slowly Changing Dimension Type 2 history

60-second recall

What strong reasoning looks like Use half-open intervals: `valid_from <= event_date < valid_to`. C001 should map O101 to East and O102 to South. Prove exactly one current row, every `valid_from < valid_to`, and no overlapping windows. If both orders report South, you joined to current state. If a fact duplicates, overlapping dimension windows are a likely cause. Without opening the lesson, explain Slowly Changing Dimension Type 2 history in 60-120 seconds. State the row/object grain, one realistic failure, one database-enforced rule (if applicable), and one independent validation

[Refresh source: TechLake - DWH Tutorial 16: Slowly Changing Dimensions and Types](#)

Recall task

Supplier risk Low->High; compare current-risk report with risk-at-receipt report.

DE07-10 | Build and validate a small analytical database

60-second recall

What strong reasoning looks like End-to-end validation should use redundant controls. The supplied source has 9 lines / 2280.00 total; the Completed population has 8 lines / 1780.00 / 5 distinct orders. The fact must match the filtered population, not the unfiltered source. A correct amount with wrong row count can hide offsetting duplicate/missing rows. A correct row count with wrong amount can hide corrupted measures. Without opening the lesson, explain Build and validate a small analytical database in 60-120 seconds. State the row/object grain, one realistic failure, one database-enforced rule (if applicable), and one independent vali

Recall task

Fresh receipt/supplier analytical database with different keys, history attribute and reconciliation controls.

DE07-11 | Bulk loading with COPY / staging boundaries

60-second recall

What strong reasoning looks like Do not confuse SQL COPY with psql \copy. Server-side COPY sees server paths/permissions; \copy makes the psql client read the file. Raw staging should preserve inspectable source values before typed promotion. For customer_stage_bad.csv, source rows=5, accepted=2, rejected/duplicate=3. Keep reject reasons. Without opening the lesson, explain Bulk loading with COPY / staging boundaries in 60-120 seconds. State the row/object grain, one realistic failure, one database-enforced rule (if applicable), and one independent validat

[Refresh source: PostgreSQL 18 documentation - COPY](#)

Recall task

Fresh dirty receipt CSV with malformed amount, duplicate line key and valid late row; preserve rejects.

DE07-12 | Schemas, roles and least-privilege boundaries

60-second recall

What strong reasoning looks like A least-privilege design separates ownership/admin, routine writes and reads. Schema USAGE and table SELECT/INSERT are separate layers. Test a denial that the policy intends - for example analyst INSERT should fail. If you cannot safely create roles on your environment, label the plan as pending instead of fabricating screenshots. Without opening the lesson, explain Schemas, roles and least-privilege boundaries in 60-120 seconds. State the row/object grain, one realistic failure, one database-enforced rule (if applicable), and one independent valid

[Refresh source: databasetorque - PostgreSQL roles, schema and security](#)

Recall task

Reporter role can read serving only; cannot see staging/write/create. List exact minimum grants.

DE07-13 | Schema change, backup/restore and handoff validation

60-second recall

What strong reasoning looks like A dump file is not recovery proof. pg_dump exports one database; cluster-global roles/tablespaces are separate concerns. Restore into a different clean database, then rerun the same object/data/business controls. A staged migration should separate adding a structure from backfill/validation and final constraint enforcement where appropriate. Without opening the lesson, explain Schema change, backup/restore and handoff validation in 60-120 seconds. State the row/object grain, one realistic failure, one database-enforced rule (if applicable), and one independent

[Refresh source: PostgreSQL 18 documentation - pg_dump](#)

Recall task

Fresh model dump/restore into a new target; eliminate one hidden absolute-path or local-state dependency.